CM3191 Chemical Synthesis Experiments

Inorganic Experiment 2: The Synthesis, Purification and Characterization of Ferrocene

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1. Aim of experiment

- To synthesize ferrocene using iron(II) chloride with cyclopentadiene under basic condition
- To purify crude ferrocene by sublimation
- To use various analytical techniques to characterize and assess the purity of synthesized ferrocene

2. Result and Discussion

2.1 Percentage yield

Percentage yield of crude ferrocene (Session 1) = 161.1% (calculations in Appendix 5.1)

Percentage yield of purified ferrocene (Session 2) = 57.02% (calculations in Appendix 5.1)

The percentage yield of crude ferrocene is significantly higher than 100%, suggesting that large quantities of by-product and water were present. This assumption was confirmed after purifying ferrocene by sublimation. The first round of sublimation at 100 °C mostly resulted in water with minimal quantities of ferrocene crystal sublimed to the petri dish cover. Substantial amount of light brown coloured by-product remained on the petri dish base after sublimation completed (Figure 1). The by-product is predicted to contain dicyclopentadiene which could have reformed via retro Diels-Alder reaction of cyclopentadiene at 25 °C[1] while being transferred to the reaction mixture. Since dicyclopentadiene is insoluble in water, it may not have been filtered during suction filtration and washing with water.

The purified ferrocene obtained by sublimation resulted in the percentage yield of 57.02%, which is considerably lower than the literature yield of 85.2%[2]. This is attributed to reagent ratio, oxidation of iron(II) and the presence of water. Makhaev et al. reported 85.2% yield using cyclopentadiene and $FeCl_2$ in 3.4:1 ratio and identified that the yield increases when cyclopentadiene is in excess[2]. Since 2:1 ratio was employed in this experiment, the low yield could be due to only using stoichiometric ratio of reagents. Dicyclopentadiene, the predict by-product, further emphasizes the importance of using cyclopentadiene in excess because of the risk of losing cyclopentadiene by the retro Diels-Alder reaction. Similarly, iron(II) oxidation may have reduced the amount of iron(II) available to produce ferrocene, which can be reverted by introducing a small quantity of iron powder for iron(III) reduction[3]. Noting that cyclopentadiene is highly moisture sensitive, using dry solvent and anhydrous $FeCl_2$ and quickly transferring KOH to avoid hydration will also help increase the yield. These practices are critical since having excess of water will prevent KOH in the aqueous phase from effectively deprotonating cyclopentadiene in the immiscible organic phase. Hence, the use of excess cyclopentadiene and efforts to minimize moisture and oxygen will lead to an increased yield.

2.2 Stereomicroscopy

As shown in Figure 2(a), crude ferrocene appears as a yellow-brown powder. The microscopic image suggests that small quantities of pure ferrocene crystals are already present, as a few small shining particles could be observed. The opaque powder appearance indicates the presence of impurities that hindered crystal growth and the establishment of long-range order. Contrastingly, Figure 2(b) displays purified ferrocene characterized by larger, transparent yellow crystals. The right angles and unequal lengths of the sides imply the monoclinic crystal structure of ferrocene[4]. However, other crystal shapes were also observed. In addition to tail growth and linear crystal fragments, the majority exhibited hexagonal shape with approximately 120° angle. The inconsistency and diversity of crystal shapes suggest interrupted crystal growth. To promote better growth of large monoclinic crystals, sublimation should be done for a longer duration without interruption.

2.3 ^{1}H NMR Spectrum

Excluding $CHCl_3$ reference peak at 7.26 ppm, three singlet peaks were observed (Figure 3). Only the peak at 4.19 ppm is attributed to ferrocene, as all protons belong to the cyclopentadienyl rings. Due to the five-fold symmetry of the ring and the delocalization of electrons, all protons are aromatic and chemically equivalent. Consequently, only one singlet peak has resulted because there are no adjacent protons with which the aromatic protons can couple. The integration was normalised to 10.00 for 4.19 ppm and thus resulted in an integration value of 10.29 for the singlet peak at 1.56 ppm which represents H_2O. Considering the integration values and the 10:2 ratio for the number of protons in ferrocene to H_2O, it can be concluded that the product consists of approximately 17 mol% of ferrocene. However, the mass fraction of H_2O in the final product is insignificant considering that the molecular weight of ferrocene (186.04 g/mol) is more than ten times that of H_2O (18.015 g/mol). The presence of the H_2O signal also implies that ferrocene was not fully dried. Prolonged sublimation at 100 °C to remove water, followed by a gradual increase in temperature, could yield water-free ferrocene. Additionally, the peak at 2.17 ppm with integration value of 0.43 corresponds to the trace amount of acetone remaining after cleaning the metal spatula and the round bottom flask used to prepare the ^{1}H NMR sample.

2.4 ATR-IR Spectrum

ATR-IR spectrum peaks are tabulated in Table 1. It was identified that staggered ferrocene with D_{5d} symmetry was produced as the wavenumbers closely matched that of the observed and theoretical calculations for the D_{5d} conformer[5]. Characteristic peaks involve 3094.25 cm^{-1} for C-H stretch, 1489.03 cm^{-1} for C=C asymmetric stretch, and 1407.21 cm^{-1} for C-C asymmetric stretch. In addition, 1104.10 cm^{-1} corresponds to asymmetric ring breathing where one cyclopentadienyl group shrinks while the other expands. Four peaks with low wavenumbers are associated with C-H bending modes where, except for CCC in-plane bending at 854.04 cm^{-1}, represent in- and out-of-plane bending of C-H bonds in cyclopentadienyl groups[6].

However, additional peaks were observed. The weak peaks at 3903.43, 3649.10 and 3750.34 cm^{-1} may indicate spectral interference. These additional peaks can be avoided by densely packing the ferrocene crystals to prevent light penetrating to the detector or using KBr pellets. Though the cluster of peaks at 1772.54, 1698.16, and 1652.48 cm^{-1} are not identified in literature results[6], they may correspond to aromatic overtone or additional peaks for aromatic C=C stretching and bending[7].

2.5 ESI-MS Spectrum

Three peaks were observed in the ESI-MS spectrum (Table 2). Based on the molecular weight of ferrocene with the most abundant ^{56}Fe isotope, the base peak at m/z = 186.04 represents the molecular ion of the parent compound, $[Fe(C_5H_5)_2]^+$, without any fragmentation. The second intense peak at m/z = 187.04 corresponds to the protonated ferrocene, $[Fe(C_5H_5)_2H]^+$, which is an adduct ion formed by the addition of a proton from the methanol carrier solution. The m/z = 184.12 peak is attributed to $[Fe(C_5H_5)_2]^+$ containing the ^{54}Fe isotope. Considering the relative abundance of iron isotopes, the intensity of m/z = 184.12 peak should be greater than that of m/z = 187.04, because ^{54}Fe (5.845%) is approximately twice as abundant as ^{57}Fe (2.119%)[8]. However, the m/z = 187.04 peak is more intense because it corresponds to two molecular ions: $[Fe(C_5H_5)_2]^+$ with ^{57}Fe and $[Fe(C_5H_5)_2H]^+$ with ^{56}Fe. Minor peaks at m/z = 185.12 and 188.04 indicate $[Fe(C_5H_5)_2H]^+$ of ^{54}Fe and ^{57}Fe, respectively, with their significantly low intensities resulting from both the low probability of protonation and considerably lower abundance compared to the ^{56}Fe isotope.

2.6 UV-Vis

The molar extinction coefficient (ε) values are 88.991 $M^{-1}cm^{-1}$ and 48.907 $M^{-1}cm^{-1}$ for 435.00 nm and 319.50 nm, respectively (calculations in Appendix 5.6). These values closely match the literature values of 91.5 $M^{-1}cm^{-1}$ at 442 nm and 51.3 $M^{-1}cm^{-1}$ at 325 nm[8]. The absorption bands represent electronic transitions occurring within ferrocene. The 435.00 nm band represents the d-d* transition within iron(II) where electron is excited and $t_{2g} \rightarrow e_g$ transition occurs[9]. Since 435.00 nm corresponds to blue light, this absorption contributes to the yellow-orange colour of the ferrocene crystal. The 319.50 nm band represents n-π* electronic transition where iron(II) donates electron to the cyclopentadienyl rings which act as electron acceptors[10]. This transition is referred to as metal ligand charge transfer or intra-molecular charge transfer transition. Both bands exhibit small ε values as they are Laporte-forbidden ligand field transitions which have a low probability of transition[9].

2.7 Magnetic susceptibility

The experimental magnetic moment was determined to be 0.7832 B.M., indicating that the number of unpaired electrons is 0.2702 (calculations in Appendix 5.7). This value slightly deviates from the predicted spin-only value of magnetic moment for ferrocene, which is zero. However, since the calculated value is approximately zero, it supports the theory that ferrocene is diamagnetic with all electrons fully paired. The purified ferrocene magnetic moment contrasts to that of $FeCl_2 \cdot 4H_2O$ with 5.25 B.M., implying that there are four unpaired electrons and $FeCl_2 \cdot 4H_2O$ is paramagnetic.

3. Conclusion

Despite the low percentage yield of 57.02% and the presence of water in the final product, ferrocene was successfully synthesized and purified with no impurities. The ^1H NMR spectrum showing the intense peak for protons of water imply that water was sublimed along with ferrocene. However, the microscopic image of monoclinic crystals, characteristic peaks at 435.00 and 319.50 nm in the UV-Vis spectrum and the magnetic susceptibility value prove that pure ferrocene was obtained.

4. References

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4. Brock, C. P.; Fu, Y., *Acta Cryst.*, **1997**, 53(6), *928-938.*
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5. Appendices

5.1 Percentage yield calculation

$$FeCl_2{\cdot}4H_2O + 2C_5H_6 + 8KOH \rightarrow Fe(C_5H_5)_2 + 2KCl + 6KOH{\cdot}H_2O$$

Moles of cyclopentadiene used: 1.0 (mL) × 0.80 (g/mL) ÷ 66.10 (g/mol) = 0.01210 mol

Moles of $FeCl_2{\cdot}4H_2O$ used: 1.3017 (g) ÷ 198.81 (g/mol) = 0.006547 mol

As 0.01210 ÷ 2 = 0.006050 mol is less than moles of $FeCl_2{\cdot}4H_2O$ used, C_5H_6 is the limiting reagent.

Theoretical yield of ferrocene: 0.01210 (mol) ÷ 2 × 186.05 (g/mol) = 1.1256 g

Percentage yield of crude ferrocene: 1.8134 (g) ÷ 1.1256 (g) × 100% = 161.1%

Percentage yield of purified ferrocene: 0.6418 (g) ÷ 1.1256 (g) × 100% = 57.02%

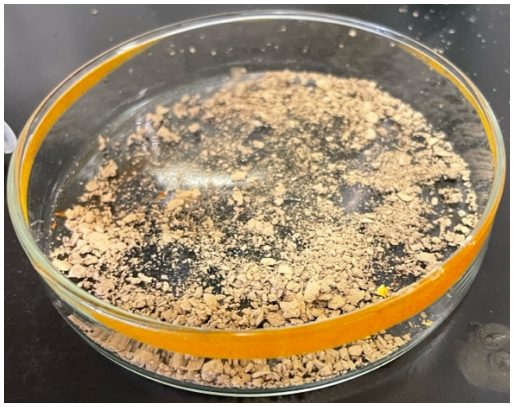


Figure 1: Image of the by-product resulted after complete sublimation

5.2 Stereomicroscopic image

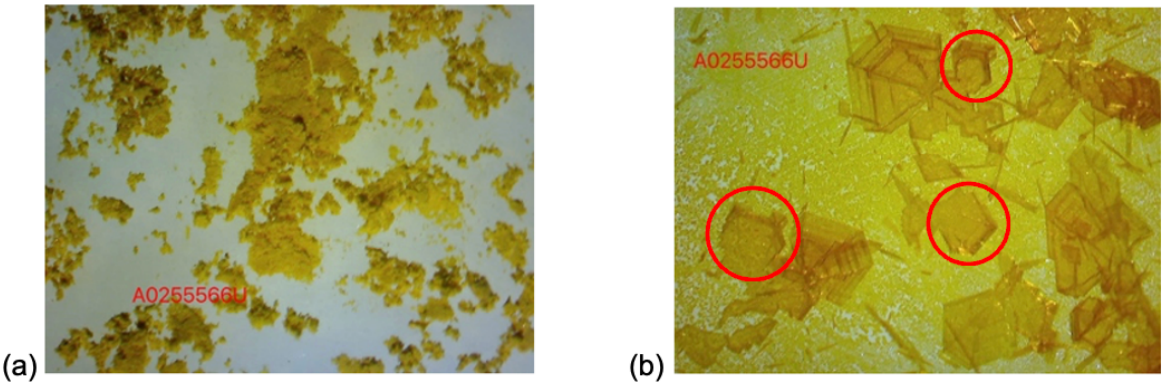


Figure 2: Microscopic images at 14x magnification ratio for (a) crude and (b) purified ferrocene with circles showing monoclinic crystals

5.3 ^{1}H NMR Spectrum

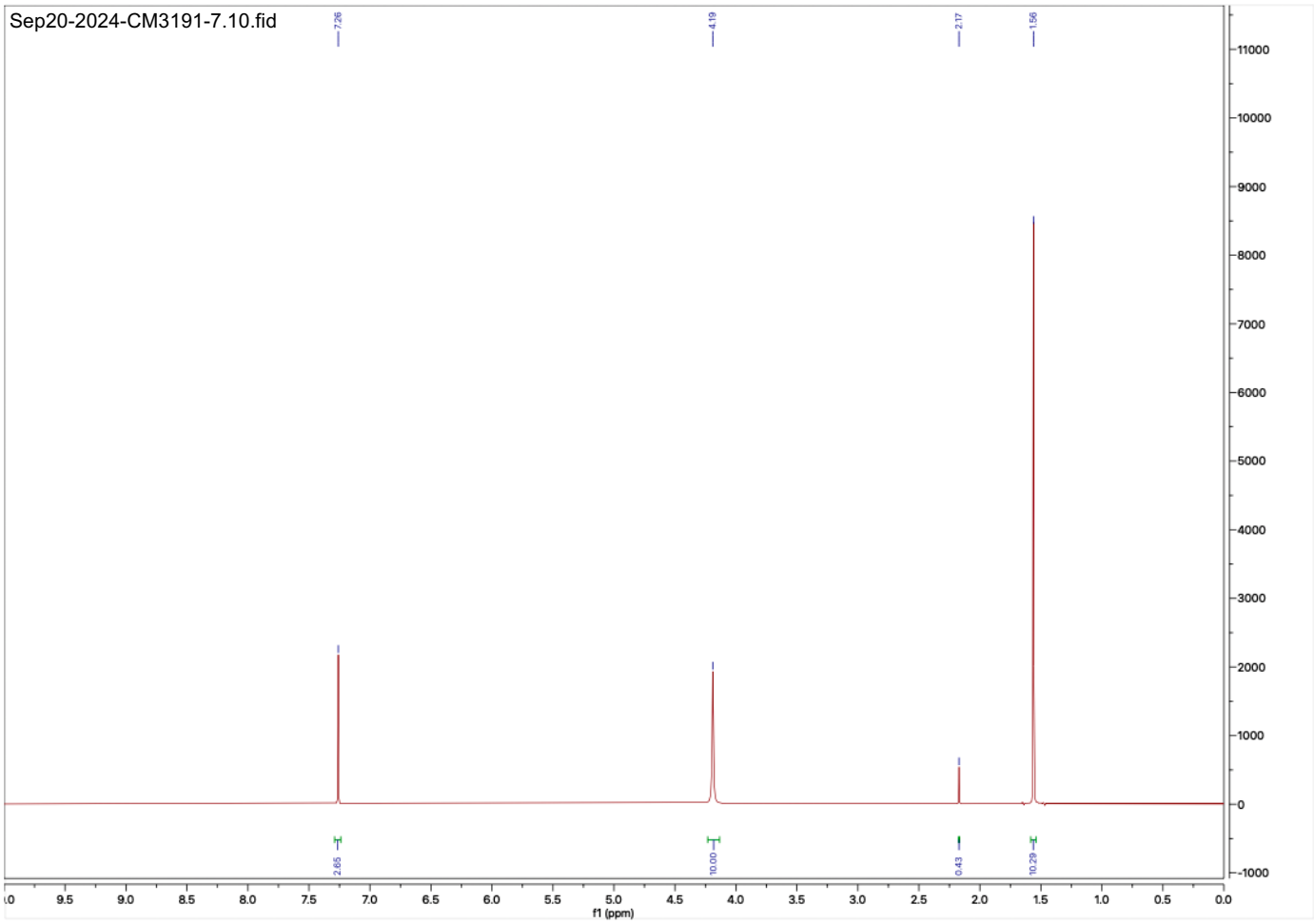


Figure 3: ^{1}H NMR spectrum of purified ferrocene (400 MHz, $CDCl_3$, 298 K)

5.4 ATR-IR Spectrum

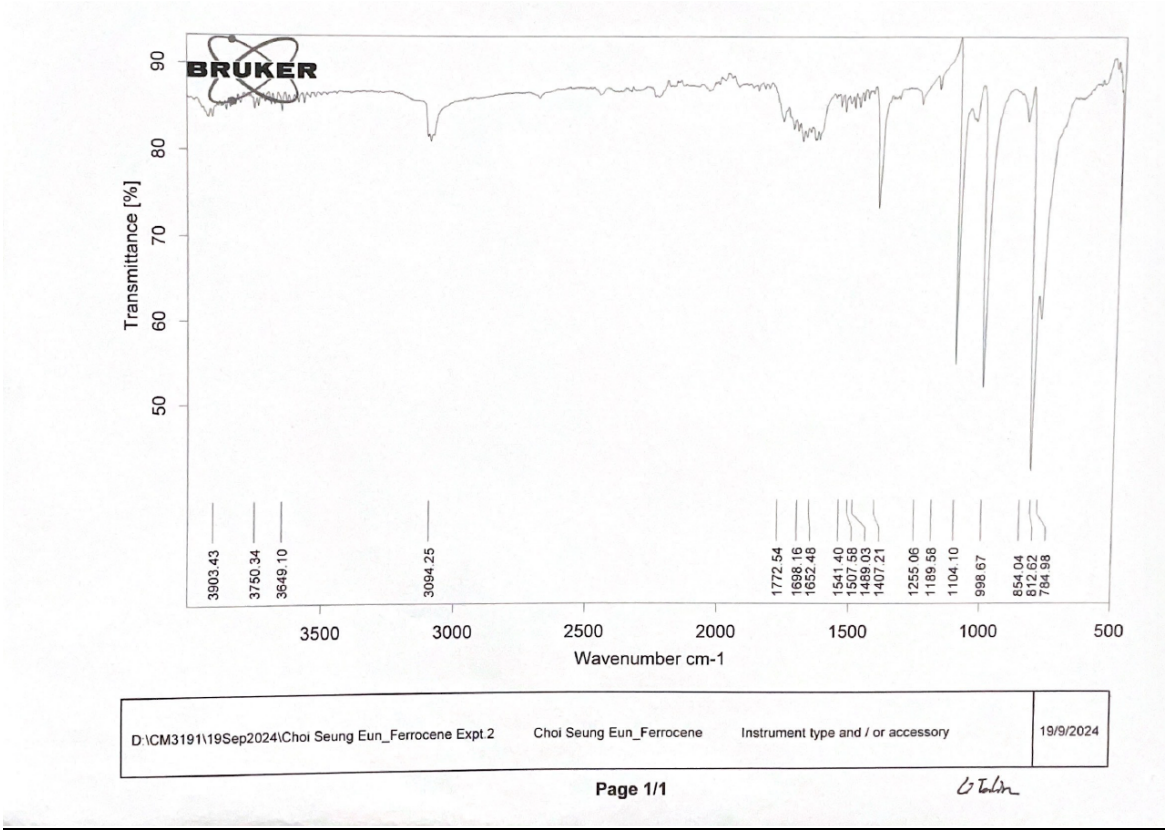


Figure 4: Scanned image of the ATR-IR spectrum of purified ferrocene

Wavenumber (cm^{-1})	Intensity	Vibrational mode
3094.25	Weak and broad	C-H stretch (A_{2u})
1772.54	Weak and broad	Aromatic overtone
1698.16	Weak and broad	Aromatic overtone
1652.48	Weak and broad	Aromatic overtone
1489.03	Weak and sharp	C=C stretch (E_{1u})
1407.21	Medium and sharp	C-C stretch (E_{1u})
1255.06	Weak and broad	CCH out-of-plane bending (A_{1u})
1104.10	Strong and sharp	Asymmetric ring breathing (A_{2u})
998.67	Strong and sharp	C-H in-plane bending (E_{1u})
854.04	Weak and sharp	CCC out-of-plane bending (E_{1u})
812.62	Strong and sharp	C-H out-of-plane bending (A_{2u})
784.98	Medium and sharp	C-H out-of-plane bending (A_{2u})

Table 1: ATR-IR spectrum peaks with respective wavenumber, intensity and vibrational mode

5.5 ESI-MS

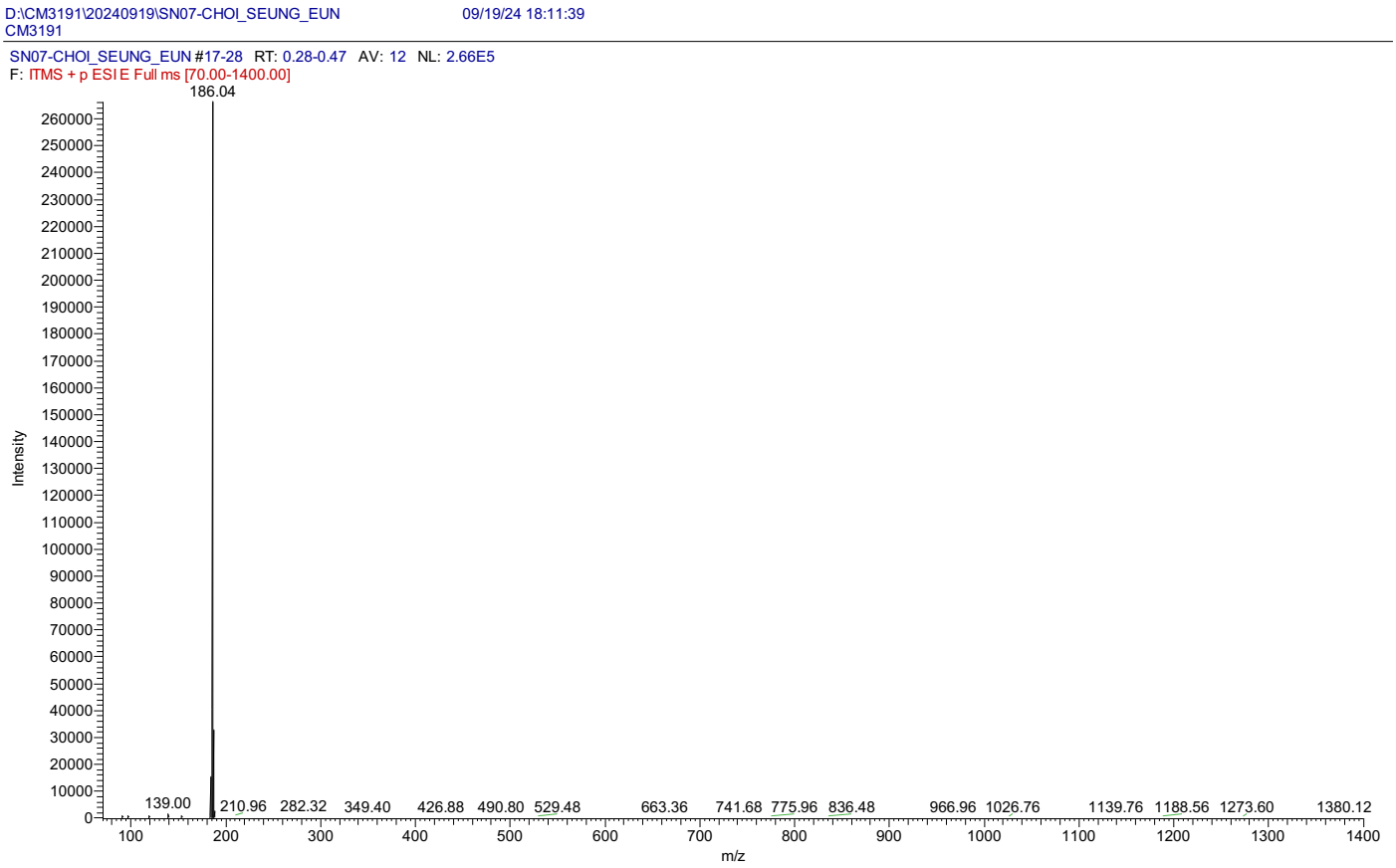


Figure 5: Full ESI-MS spectrum of purified ferrocene

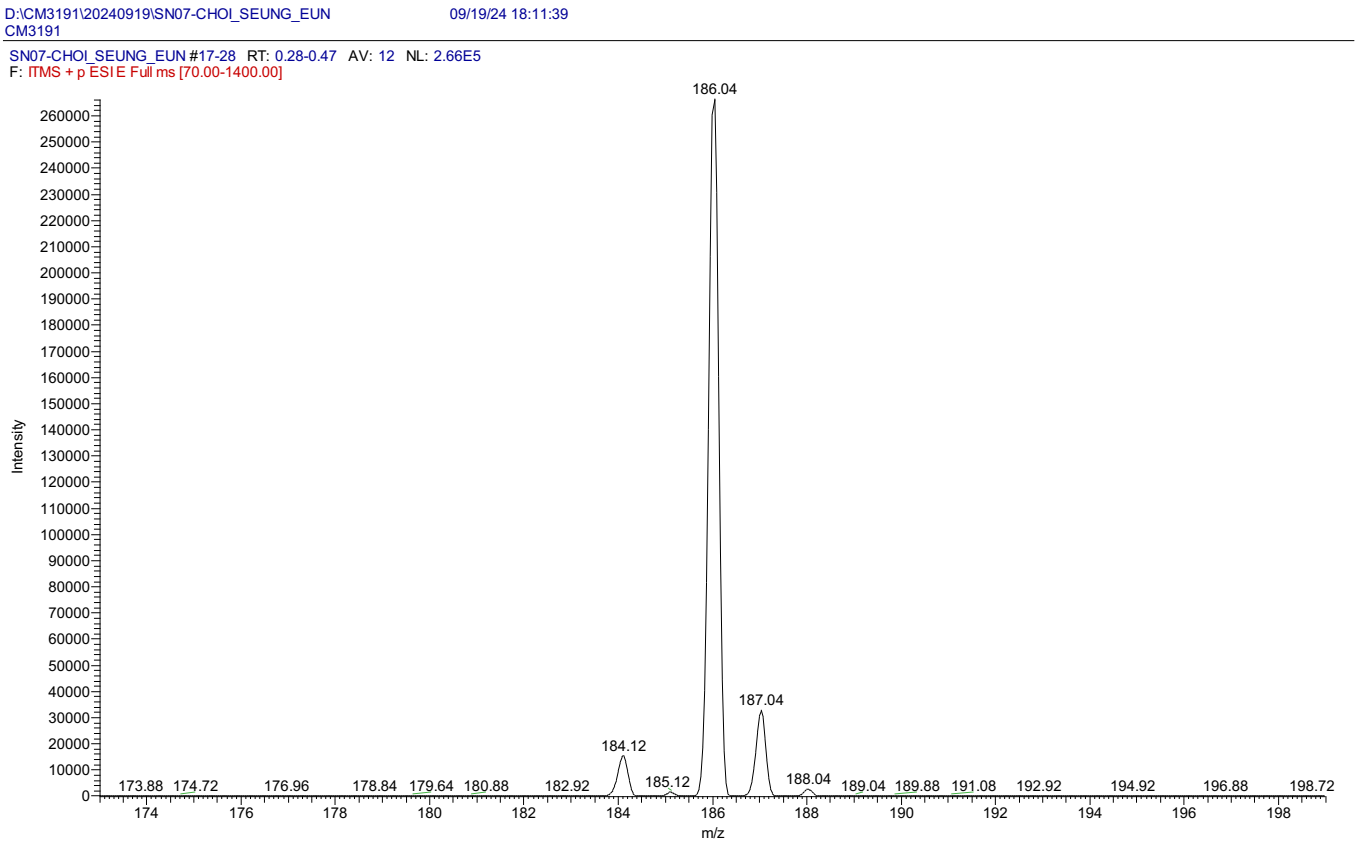


Figure 6: Zoomed ESI-MS spectrum of purified ferrocene

m/z	Intensity	Relative intensity (%)
184.12	16000	5.97
186.04	268000	100
187.04	34000	12.69

Table 2: ESI-MS spectrum peaks with respective m/z, intensity and relative abundance values

5.6 UV-Vis

Molar concentration of ferrocene solution: 0.0369 (g) ÷ 186.05 (g/mol) ÷ 25×10^{-3} (L) = 7.9334×10^{-3} M

Molar extinction coefficient at 435.00 nm: 0.706 ÷ 7.9334×10^{-3} (M) ÷ 1 (cm) = 88.991 $M^{-1}cm^{-1}$

Molar extinction coefficient at 319.50 nm: 0.388 ÷ 7.9334×10^{-3} (M) ÷ 1 (cm) = 48.907 $M^{-1}cm^{-1}$

Wavelength (nm)	Absorbance	Molar extinction coefficient ($M^{-1}cm^{-1}$)
435.00	0.706	88.991
319.50	0.388	48.907

Table 3: UV-Vis spectrum peaks with respective wavelength, absorbance and molar extinction coefficient

5.7 Magnetic susceptibility Calculation

Molar susceptibility: χ_M(total) = χ_g(total) × M.W. = 0 ($ergG^{-1}g^{-1}$) × 186.04 (g/mol) = 0 $ergG^{-1}mol^{-1}$

Correction for diamagnetic contributions for C_5H_5: -65×10^{-6} × 4 = -2.60×10^{-4} $ergG^{-1}mol^{-1}$

Molar susceptibility of paramagnetic atom: χ_a = χ para = 0 - (-2.60×10^{-4}) = 2.60×10^{-4} $ergG^{-1}mol^{-1}$

Experimental magnetic moment: μ_{eff} = 2.828 × [2.60×10^{-4} ($ergG^{-1}mol^{-1}$) × 295]$^{1/2}$ = 0.7832 B.M.

Number of unpaired electrons:

0.7832 = $[n\times(n+2)]^{1/2}$

$n\times(n+2) = 0.6134$

n = 0.2702

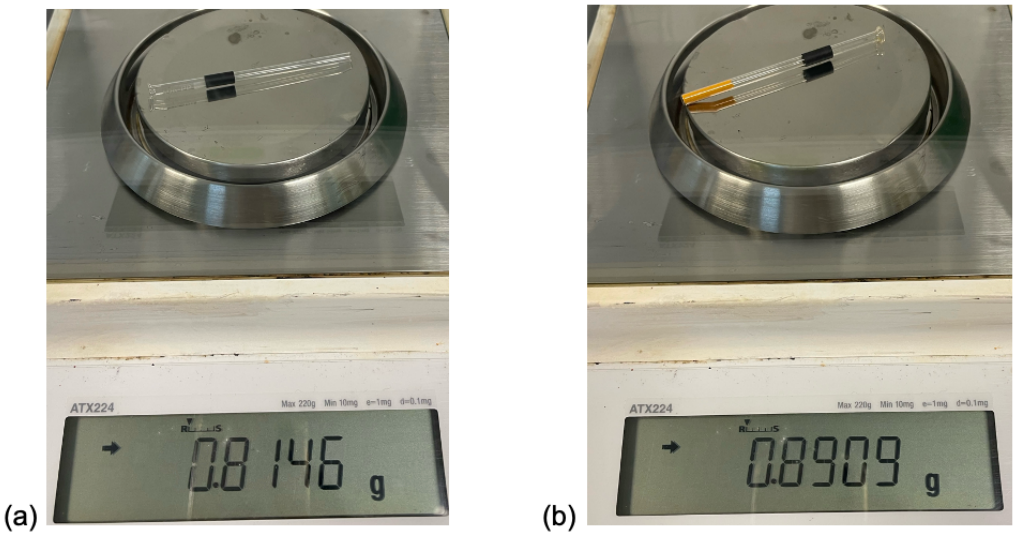


Figure 7: Image of the masses of the (a) empty and (b) filled sample tube

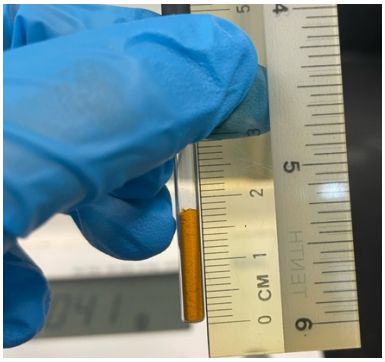


Figure 8: Image of the length of the filled sample tube

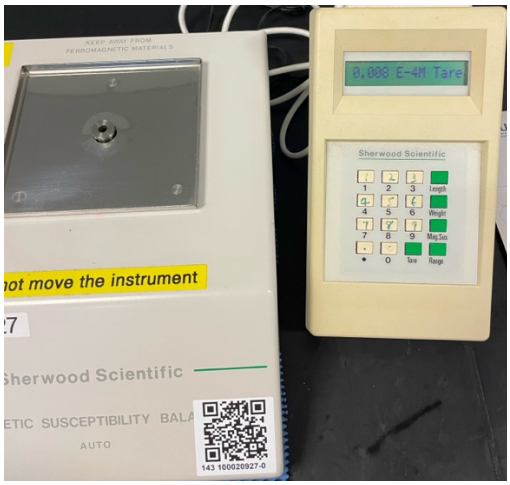


Figure 9: Image of the control unit with zero value obtained (0.000 E-4M Tare)



Figure 10: Image of the control unit with zero value obtained (0.000 E-4M Tare) for the empty sample tube

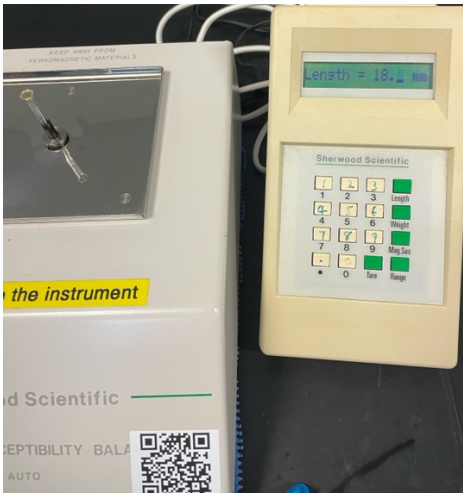


Figure 11: Image of the control unit with the filled sample tube length entered



Figure 12: Image of the control unit with the weight of the filled sample tube entered

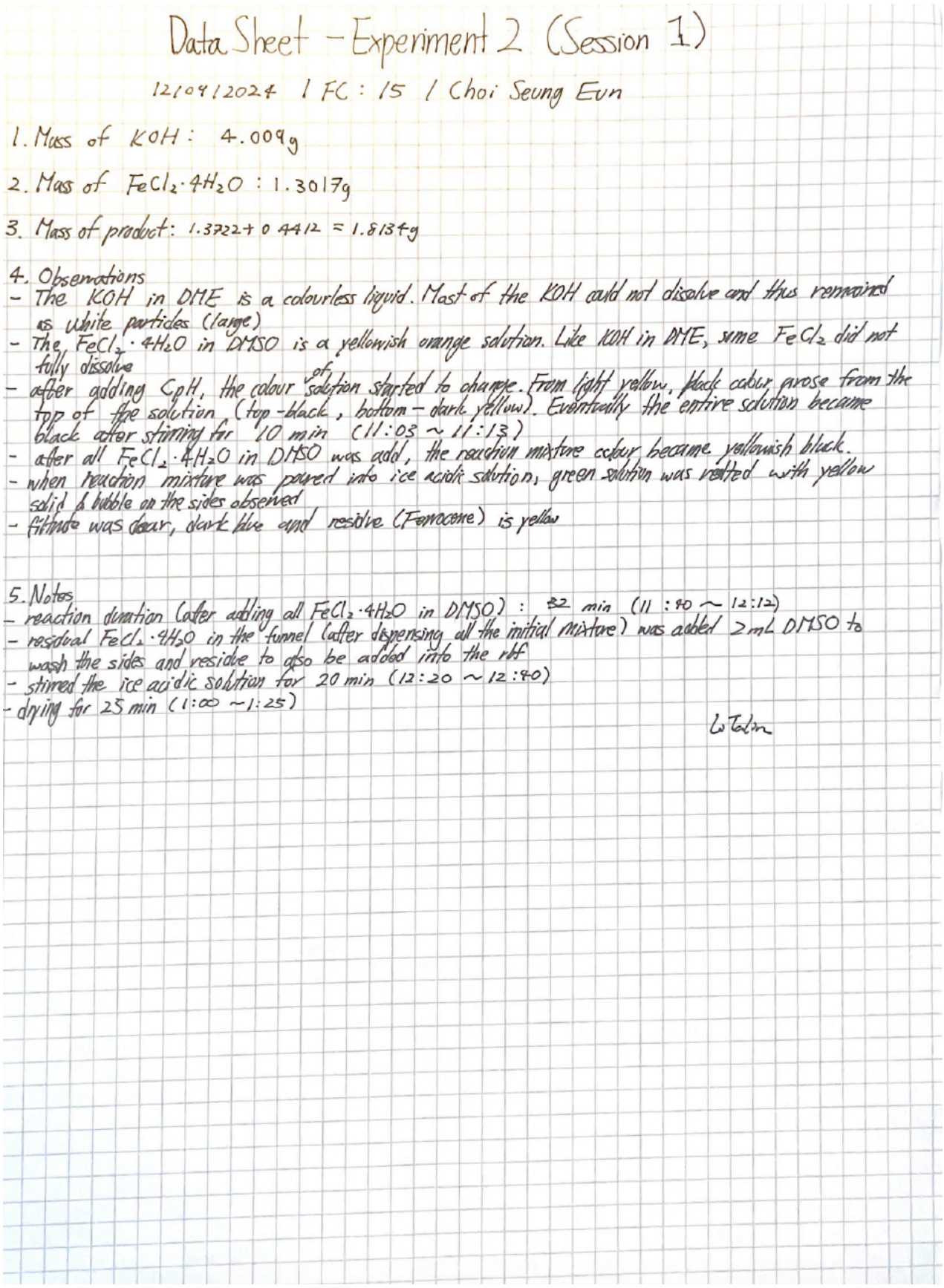


Figure 13: Image of the control unit with the volume susceptibility displayed



Figure 14: Image of the control unit with the mass susceptibility displayed

5.8 Data Sheet



Data Sheet – Experiment 2 (Session 1)

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1. Mass of KOH : 4.009g

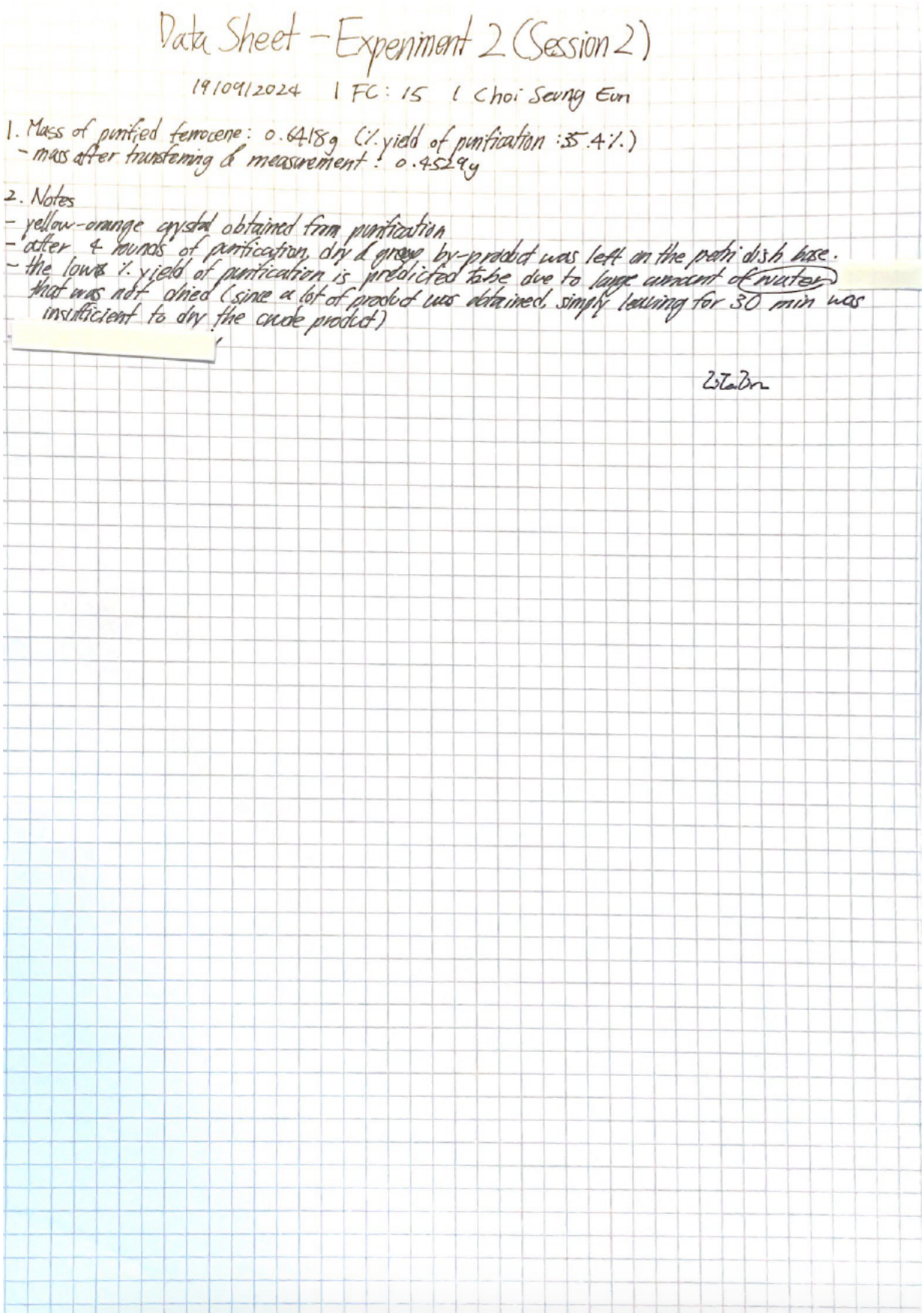
2. Mass of $FeCl_2 \cdot 4H_2O$: 1.3017g

3. Mass of product: 1.3722 + 0.4412 = 1.8134g

4. Observations
- The KOH in DME is a colourless liquid. Most of the KOH could not dissolve and thus remained as white particles (large)
- The $FeCl_2 \cdot 4H_2O$ in DMSO is a yellowish orange solution. Like KOH in DME, some $FeCl_2$ did not fully dissolve
- after adding C_5H_6, the colour of solution started to change. From light yellow, black colour arose from the top of the solution (top - black, bottom - dark yellow). Eventually the entire solution became black after stirring for 10 min (11:03 ~ 11:13)
- after all $FeCl_2 \cdot 4H_2O$ in DMSO was add, the reaction mixture colour became yellowish black.
- when reaction mixture was poured into ice acidic solution, green solution was melted with yellow solid & bubble on the sides observed
- filtrate was clear, dark blue and residue (Ferrocene) is yellow

5. Notes
- reaction duration (after adding all $FeCl_2 \cdot 4H_2O$ in DMSO) : 32 min (11:40 ~ 12:12)
- residual $FeCl_2 \cdot 4H_2O$ in the funnel (after dispensing all the initial mixture) was added 2mL DMSO to wash the sides and residue to also be added into the rbf
- stirred the ice acidic solution for 20 min (12:20 ~ 12:40)
- drying for 25 min (1:00 ~ 1:25)

Figure 15: The scanned data sheet for session 1


Data Sheet - Experiment 2 (Session 2)

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1. Mass of purified ferrocene: 0.6418g (% yield of purification: 35.4%)
- mass after transferring & measurement: 0.4529g

2. Notes
- yellow-orange crystal obtained from purification
- after 4 rounds of purification, dry & greasy by-product was left on the petri dish base.
- the lower % yield of purification is predicted to be due to large amount of water that was not dried (since a lot of product was obtained, simply leaving for 30 min was insufficient to dry the crude product)

Figure 16: The scanned data sheet for session 2